

Uno R4 Minima → DFRobot DFR0669 TFT · LiPower battery shield

DFRobot DFR0669 (ILI9488, 480x320, 3.3-5.5 V) · GT911 touch I2C · LiPower shield (3.7 V LiPo → 5 V)

AA-30.ZERO: UART1 TX → R4 D0 (Serial1 RX) · UART1 RX → R4 D1 (Serial1 TX) · GND common.

Module runs at 3.3-5.5 V, so SPI logic needs NO level shifter (earlier 2.4" ILI9341 + LEVEL-8P + HU1-HU8 are removed). Backlight (BL) is on by default.

Display uses SPI D8-D13; touch uses I2C A4/A5; AA-30 uses D0/D1 — no overlap. Touch INT and \$DCS are not used.

Display: SPI D8-D13 · touch: I2C A4/A5 · AA-30: D0/D1 — no overlap. MISO optional.

LiPower: 5 V powers R4 + DFR0669; MAX17043 gauge shares I2C A4/A5 (0x36) with GT911 (0x5D); low-batt ALRT on D2.

LiPower Shield 0.5A (power + fuel gauge)

3.7 V LiPo → 5 V · MAX17043 gauge → I2C A4/A5 @0x36

low-battery ALRT → D2 · charges LiPo via mini-USB (500 mA)

